

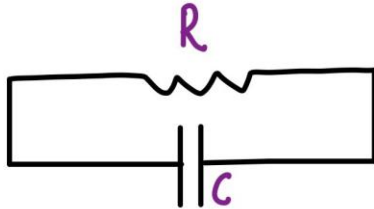
Natural Response

There are two types of Responses :

1. Natural Response
2. Forced Response

The natural response of RC and LC circuits:

a) RC



1. Write the KVL/KCL Equation for the circuit.

$$V_C - V_R = 0$$

As we know the voltage across the capacitor, $V_C = V_0$

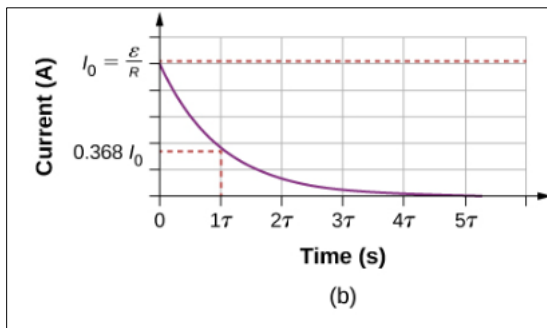
$$V_C = \frac{1}{C} \int i \, dt$$

$$\frac{1}{C} \int i \, dt - iR = 0$$

2. Reduce or convert the governing equation to a homogenous differential equation :

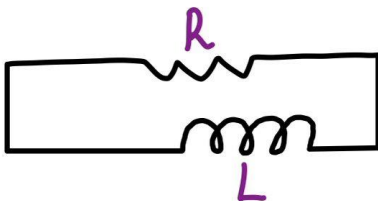
$$\frac{i}{C} - \frac{di}{dt} R = 0$$

3. Use the template solution $i = Ae^{st}$ and plug it into the homogenous equation and find the constants A & S.



* This is for both charging and discharging

b) LC



1. We can write the equation as $L \frac{di}{dt} - v_0 + \frac{1}{C} \int i \, dt$

2. Differentiate to get second order $L \frac{d^2 i}{dt^2} + \frac{i}{C} = 0$

We will use the following Equations for solving part (a) and (b).

$$(1) \quad V_C = \frac{1}{C} \int i \, dt$$

$$(2) \quad V_R = iR$$

$$(3) \quad V_L = L \frac{di}{dt}$$

An equation's order depends on the number of energy storage elements in the circuit. This also means adding power dissipating elements doesn't change the order.

3. Use template solution $i = Ae^{st}$ (assuming one solution)

$$Ls^2 Ae^{st} + \frac{1}{C} Ae^{st} = 0$$

$$Ae^{st} \left(Ls^2 + \frac{1}{C} \right) = 0,$$

$Ls^2 + \frac{1}{C}$ is known as the characteristic equation

We obtain two solutions for (s) as the equation is second order.

$$s_1 = +j\sqrt{\frac{1}{LC}} \quad s_2 = -j\sqrt{\frac{1}{LC}}$$

The root can be written as $\omega_0 = \sqrt{\frac{1}{LC}}$

$$s_1 = +j\omega_0, \quad s_2 = -j\omega_0$$

4. Now use the template $i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$ for second order equations.

$$i(t) = A_1 e^{+j\omega_0 t} + A_2 e^{-j\omega_0 t}$$

We know the general expansion for e^{jx} from Euler's Identity $e^{jx} = \cos x + j \sin x$

$$i(t) = A_1 (\cos \omega_0 t + j \sin \omega_0 t) + A_2 (\cos \omega_0 t - j \sin \omega_0 t)$$

$$i(t) = (A_1 + A_2) \cos \omega_0 t + j(A_1 - A_2) \sin \omega_0 t$$

Let $B_1 = (A_1 + A_2)$ and $B_2 = (A_1 - A_2)$

$$i(t) = B_1 \cos \omega_0 t + B_2 \sin \omega_0 t$$

5. To find B_1 and B_2 use the initial conditions of the circuit at $t = 0$

$$i(0^-) = i(0^+) = 0$$

$$v_C(0^-) = v_C(0^+) = v_0$$

We need to find $\frac{di}{dt}(0^+)$ so we look at the voltage across the inductor.

$$v = L \frac{di}{dt}$$

$$\frac{di}{dt}(0^+) = \frac{v_0}{L}$$

Apply the first initial condition (1) and get

$$0 = B_1 \cos \omega_0(0) + B_2 \sin \omega_0(0)$$

$$B_1 = 0$$

We get $i(t) = B_2 \sin \omega_0 t$

Now apply the second initial condition

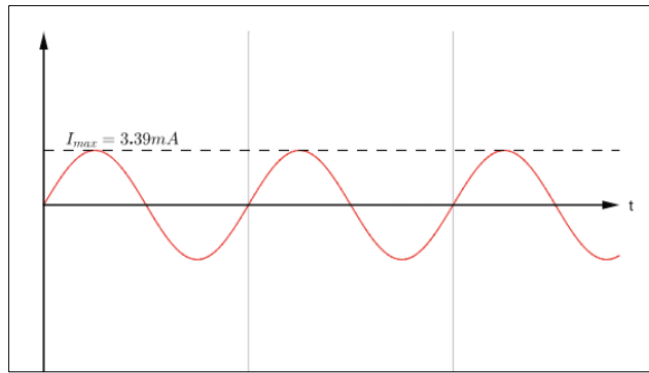
$$\left. \frac{\partial i}{\partial t} \right|_{t=0} = \frac{d}{dt} (B_2 \sin \omega_0 t) = \omega_0 B_2 \cos \omega_0 t$$

At $t=0$, $\frac{v_0}{L} = \omega_0 B_2 \cos \omega_0(0) = \omega_0 B_2$

$$B_2 = \frac{1}{\omega_0 L} v_0 = \sqrt{\frac{C}{L}} v_0$$

Finally, we get the equation:

$$i(t) = \sqrt{\frac{C}{L}} v_0 \sin \omega_0 t$$



forever (without damping).

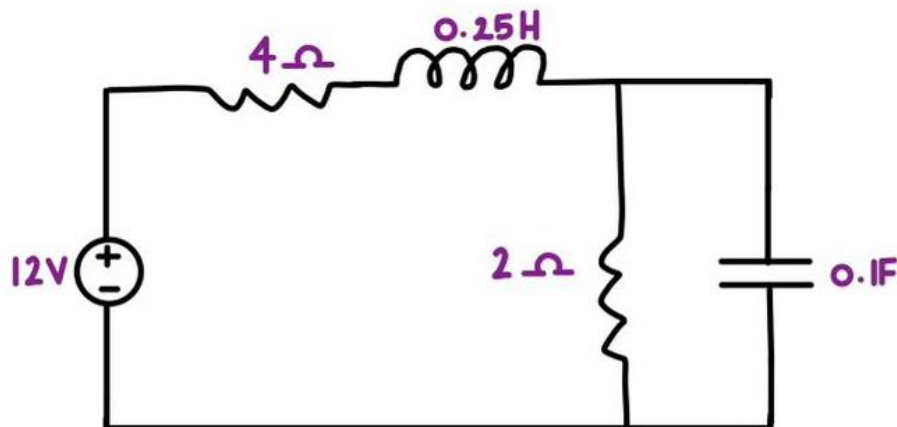
*Ideally, the graph will continue oscillating

Lecture 2 (11/2/25)

problem-solving (steady and transient state)

Q1: same as Q2 from practice problems

Q2:



(a) $i(0^+), v(0^+)$

(b) $\frac{di}{dt}(0^+), \frac{dv}{dt}(0^+)$

(c) $i(\infty), v(\infty)$

(a) Let's look at the circuit at $t = 0^-$. We short-circuit the inductor and open-circuit the capacitor.

Inductor current cannot change instantly so $i(0^+) = i(0^-) = \frac{v_0}{R_{eq}} = \frac{12}{4+2} = 2A$

Capacitor voltage cannot change instantly so $v(0^+) = v(0^-) = iR = 2A \times 2\Omega = 4V$

(b) We need to find the change in current and voltage after the circuit is opened. We can short circuit the 2Ω resistor. Let's write the KVL equation at $t = 0^+$ we have

$$v_0 = i(0^+)R + L \frac{di}{dt}(0^+) + v(0^+)$$

$$12V = 2A(4\Omega) + 0.25 \frac{di}{dt}(0^+) + 4V$$

Solve and we get the change in current:

$$\frac{di}{dt}(0^+) = 0$$

The change in voltage across the capacitor will be:

$$i(0^+) = C \frac{dv}{dt}(0^+)$$

$$\frac{dv}{dt}(0^+) = \frac{2A}{0.1F} = 20 V/sec$$

(this is correct, copied from lecture notes)

(c) When the switch is kept open for a long time the circuit goes to DC steady state

Therefore $i(\infty) = 0$ and $v(\infty) = 12V$

(RLC) Second order circuit

Steps to solve the circuit

1) Write the KVL equation

$$\begin{aligned}
 L \frac{di}{dt} + V_C + iR &= 0 \\
 L \frac{d^2 i}{dt^2} + \frac{dV_C}{dt} + \frac{di}{dt} R &= 0 \\
 L \frac{d^2 i}{dt^2} + \frac{i}{C} + R \frac{di}{dt} &= 0 \\
 \frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i &= 0
 \end{aligned}$$

This is a second-order linear homogeneous ordinary differential equation.

2) Write the template solution $i = Ae^{st}$

$$\begin{aligned}
 As^2 e^{st} + \frac{R}{L} A s e^{st} + \frac{1}{LC} A e^{st} &= 0 \\
 A e^{st} \left(s^2 + \frac{R}{L} s + \frac{1}{LC} \right) &= 0 \\
 s^2 + \frac{R}{L} s + \frac{1}{LC} &= 0
 \end{aligned}$$

Applying the quadratic formula, the two values of s we get are

$$\begin{aligned}
 s &= \frac{1}{2L} \cdot (-R \pm \sqrt{R^2 C - 4L}) \\
 s_1 &= -\alpha + \sqrt{\alpha^2 - \omega_0^2} \\
 s_2 &= -\alpha - \sqrt{\alpha^2 - \omega_0^2}
 \end{aligned}$$

where $\alpha = \frac{R}{2L}$, $\omega_0 = \frac{1}{\sqrt{LC}}$

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

3) Find A_1 and A_2 using initial conditions

$$i(0^+), \frac{di(0^+)}{dt}$$

3 types of cases are possible:

1) Roots are real and distinct (overdamped)

$$\begin{aligned}
 \alpha^2 &> \omega_0^2 \\
 i(t) &= A_1 e^{s_1 t} + A_2 e^{s_2 t}
 \end{aligned}$$

2) Roots are real and equal (critically damped)

$$\begin{aligned}
 \alpha^2 &= \omega_0^2 \\
 i(t) &= (A_1 + A_2 t) e^{-\alpha t}
 \end{aligned}$$

3) Roots are in a complex plane (underdamped)

$$\begin{aligned}
 \alpha^2 &< \omega_0^2 \\
 i(t) &= e^{-\alpha t} [B_1 \cos \omega_d t + B_2 \sin \omega_d t]
 \end{aligned}$$

1st case

$R=4 \Omega$, $L=1 \text{ H}$, $C=1/3$

$$\begin{aligned}
 \therefore \alpha &= 2, \omega_0 = \sqrt{3} \\
 s_1 &= -2 + \sqrt{4 - 3} = -2 + 1 = -1; s_2 = -2 - \sqrt{4 - 3} = -2 - 1 = -3.
 \end{aligned}$$

Since $\alpha^2 > \omega_0^2$ The roots are real and distinct.

Template solution

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

using s_1 & s_2 values,

$$i(t) = A_1 e^{-t} + A_2 e^{-3t}$$

Initial conditions:

$$1) V_C(0^+) = V_0$$

$$2) i_L(0^+) = 0$$

$$\text{Using the 1st condition; } 0 = A_1 + A_2$$

$$\therefore A_1 = -A_2$$

using the 2nd condition :

first write KVL :

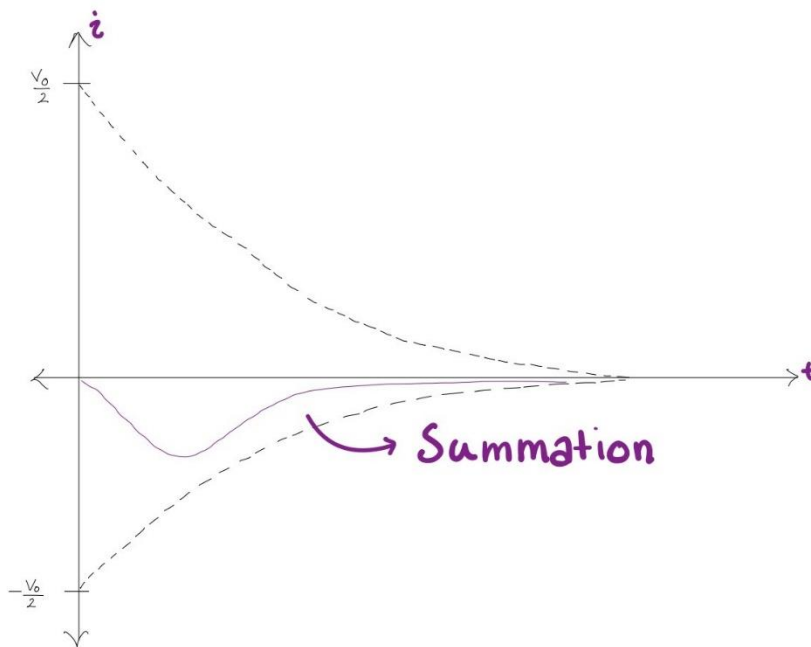
$$Ri + L \frac{di}{dt} + V_C = 0$$

$$Ri(0) + L[-A_1 e^{-t} - 3A_2 e^{-3t}] + V_C = 0$$

$$\therefore V_0 = A_1 + 3A_2$$

$$\therefore A_1 = \frac{-V_0}{2}, A_2 = \frac{V_0}{2}$$

$$\therefore i(t) = \frac{1}{2}(-V_0 e^{-t} + V_0 e^{-3t})$$



2nd case:

$$\alpha^2 = \omega_0^2$$

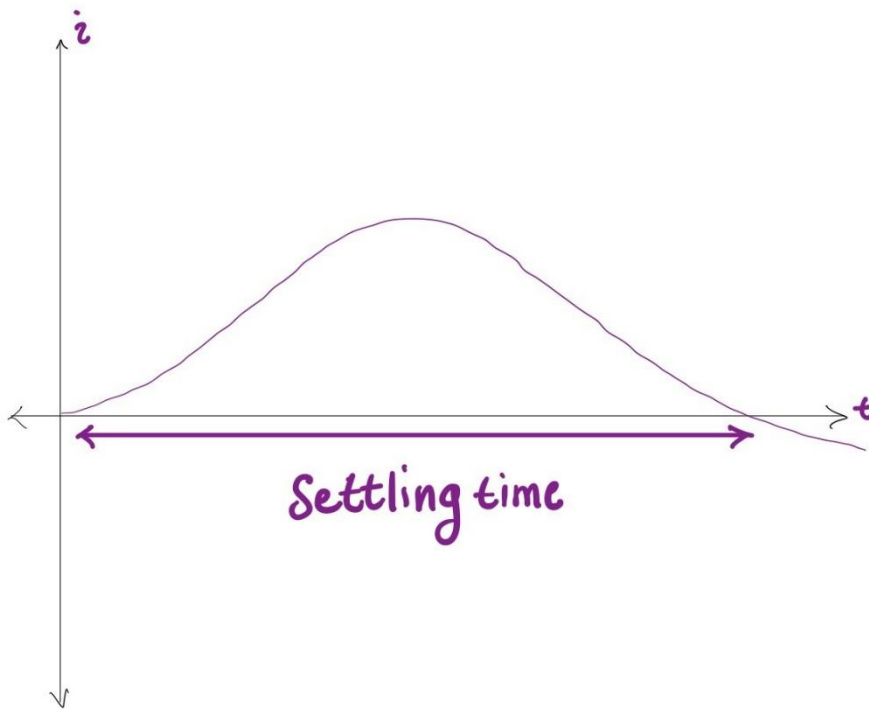
$$\therefore R^2 = \frac{4L}{C}$$

$$s_1 = s_2 = -\alpha = -$$

$$i(t) = A_1 e^{-\alpha t} + A_2 e^{-\alpha t} = e^{-\alpha t}(A_1 + A_2) = A e^{-\alpha t},$$

this does not give 2 distinct solutions,

$\therefore$ the method of reduction of order is used and the solution is; $i(t) = (A_1 + A_2 t) e^{-\alpha t}$



3rd case:

$$L = 1 \text{ H}, R = 2 \Omega, C = 1/17$$

$$\alpha = \frac{R}{2L} = \frac{2}{2} = 1$$

$$\omega_0 = \frac{1}{\sqrt{\frac{1}{17}}} = \sqrt{17}$$

* Note: In electrical engineering, j is used instead of i for imaginary numbers to avoid confusion with electrical current

$$s_1 = -1 + 4j = -\alpha + j\omega_d$$

$$s_2 = -1 - 4j = -\alpha - j\omega_d$$

Template solution

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

using s_1 & s_2 values,

$$i(t) = e^{-\alpha t} (A_1 e^{j\omega_d t} + A_2 e^{-j\omega_d t})$$

It is known that,

$$e^{j\theta} = \cos \theta + j \sin \theta$$

Therefore,

$$i(t) = e^{-\alpha t} [A_1 (\cos \omega_d t + j \sin \omega_d t) + A_2 (\cos \omega_d t - j \sin \omega_d t)]$$

$$i(t) = e^{-\alpha t} [B_1 \cos \omega_d t + B_2 \sin \omega_d t]$$

Where $B_1 = A_1 + A_2$, $B_2 = A_1 - A_2$.

$i(t)$ can also be written as:

$$i(t) = e^{-\alpha t} A \sin(\omega_d t + \theta)$$

$$i(t) = A e^{-\alpha t} \sin(4t + \theta)$$

Initial conditions

1) $i(0) = 0$

2) $V_c = V_0$

using the first condition:

$$i(0) = 0, A \sin \theta = 0$$
$$\therefore \theta = 0$$

Using the second condition :

First write KVL:

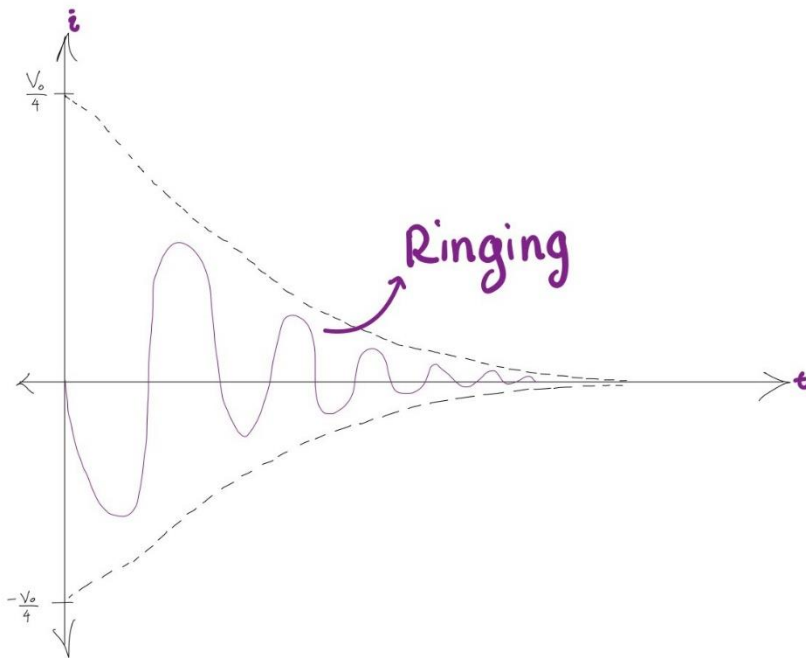
$$Ri(0) + L \frac{di(0)}{dt} + V_0 = 0$$

$$\frac{di(0)}{dt} = -V_0$$

$$4A = -V_0$$

$$A = \frac{-V_0}{4}$$

$$i(t) = -\frac{V_0}{4} e^{-t} \sin 4t$$



Example questions :

(1)LR Circuits

1. Switch S (Fig 1.a) is arranged to disconnect the source and simultaneously, at time $t=0$, short-circuit the coil of **2H** inductance and **10 ohm** resistance. If the initial current in the coil, I_0 , is **20A**, predict the current i after **0.2s** has elapsed. How long will it take for the natural behaviour of current to become zero?

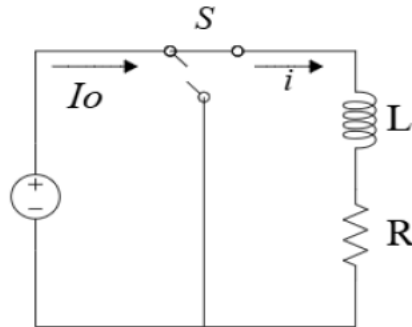


Fig 1.a

Solution:

- a. Using the template solution, we write

$$i = I_0 \cdot e^{-t/\tau} \text{ (A)}$$
- b. τ (time constant) = L/R (s)
- c. The energy stored in an inductor cannot change instantaneously.
 Therefore, the current in the coil after the switch is thrown must equal to the current just before.
- d. Calculate the values of $I(t)$ after substituting the values in the template solution

Calculations:

$$i = I_0 = 20A \text{ (At } t = 0 + \text{)}$$

$$\tau = L/R = 2/10 = 0.2 \text{ s}$$

$$I = I_0 \cdot e^{-t/\tau} = 20 \cdot e^{-t/0.2}$$

Given $t = 0.2 \text{ s}$,

Hence the solution is

$$i = 20 \cdot e^{-0.2/0.2} =$$

$$20 \times 0.368 = 7.36 \text{ A}$$

****The current decreases continually (exponential decay), but never**

becomes zero. (Theoretically and practically).

2. The switch in the circuit of Fig. 1.b has been closed for a long time. At $t=0$, the switch is opened. Calculate $i(t)$ for $t > 0$

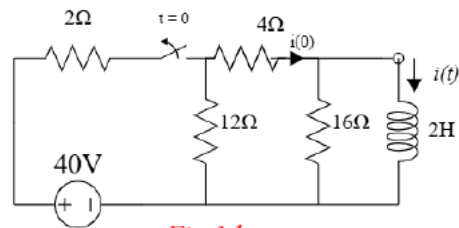


Fig 1.b

Solution:

- a. Using the template solution, we write

$$i = i_0 \cdot e^{-t/\tau} \text{ (A)}$$

- b. τ (time constant) = L/R (s)

- c. To calculate R_{th} , find effective resistance when $t = 0^+$

- d. Since the switch is closed for a long time, voltage across the inductor is zero ($di/dt = 0$), and is short circuited. Along with the inductor, the 16 ohm resistance is short circuited as well.

- e. When switch is turned off, current through the inductor cannot change instantaneously, therefore,
[The current through the 4 ohm resistor will be the same as $i(t)$ at $t = 0^-$]
This will be our I_0 .

Calculations:

To calculate R_{th} , $(4 + 12) \parallel 16$

$R_{th} = 8 \text{ ohm}$

$$= L/R_{th} = 2/8 = 1/4 \text{ s}$$

When $t < 0$:

$$R_{eff} = 4 \parallel 12 + 2 = 5 \text{ ohm}$$

$$I(\text{total}) = 40/5 = 8 \text{ A}$$

Using the concept of current divider, the current through 4 ohm resistor is-

$$8 \times \frac{12}{12 + 4} = 6 \text{ A}$$

Therefore,

$$i_0 = i(t)(0^-) = 6 \text{ A}$$

Substituting in template solution, $t > 0$

The solution is,

$$i(t) = 6e^{-4t} \text{ A}$$

For this question, you can refer to Fundamentals of electric circuits (4th edition) , Example 7.4, Pg 263, Chapter 7

(2)RLC Circuits (Series)

The study of natural response of RLC circuits is a necessary background for studies in filter design and communication networks. Let us practice some basic problems to grasp the concept well.

1. If $R = 10 \text{ ohm}$, $L = 5\text{H}$ and $C = 2\text{mF}$, find α , ω , s_1 and s_2 .

What type of natural response will the circuit have?

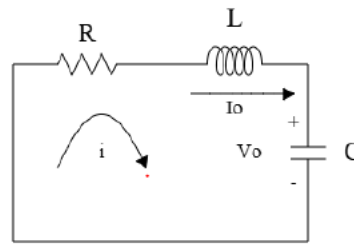


Fig 2. a

Solution:

- a. We know the formula of $\alpha = R/2L$. We substitute the values of L and R to obtain the same
- b. Formula of $\omega = 1/\sqrt{LC}$
- c. Template solution for s_1 and s_2 is:
$$-\alpha (+-) \sqrt{\alpha^2 - \omega^2}$$
- d. To find out what kind of natural response the circuit will have
 $\alpha > \omega$; (*underdamped*)
 $\alpha = \omega$; (critically damped)
 $\alpha < \omega$; (*overdamped*)

Calculations:

Substituting values of R and L we get

$$\alpha = 1$$

Substituting the values of L and C we get

$$\omega = 10$$

Using the template solution:

$$s_1 = -1 + j9.95$$

$$s_2 = -1 - j9.95$$

Since $\alpha < \omega$:

It is a case of underdamped.

For this question, you can refer to Fundamentals of electric circuits (4th edition) , Practice problem 8.3, Pg 324, Chapter 8

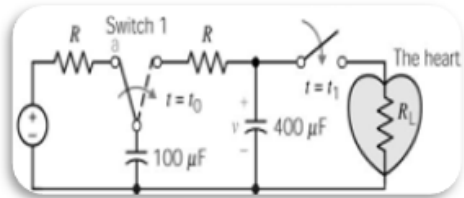
Applications in Real Life

An RC circuit has a capacitor connected with a resistor and a power source. Once the capacitor starts discharging, its charges flow through the circuit, and the resistor slows down its movement. The discharging process dissipates electrical potential energy stored in the capacitor.

Strobe Light



Pacemaker



Flash camera



An RL circuit has an inductor connected with a resistor and a power source. When the current changes, the inductor resists the change by generating a counter electromotive force (emf). The resistor dissipates energy as heat, while the inductor temporarily stores energy in its magnetic field before releasing it back into the circuit.

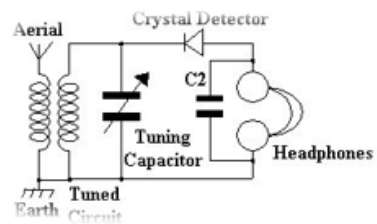
Radio Frequency Amplifier



Choke (in Tube light)



Radio wave Transmitter



An LC circuit consists of an inductor and a capacitor connected. When the capacitor discharges, the stored electrical energy is converted into a magnetic field in the inductor. The energy oscillates back and forth between the capacitor and inductor, forming an oscillatory system with minimal energy loss in an ideal case.

Proximity Card



Walkie-Talkie



Receiver Dish



An RLC circuit consists of a resistor, an inductor, and a capacitor connected. The capacitor and inductor work together to create oscillations by exchanging energy, while the resistor dissipates some of this energy as heat, gradually reducing the oscillation amplitude. The circuit's behavior depends on the balance between resistance, inductance, and capacitance, leading to overdamped, underdamped, or critically damped responses.

Band Pass Filter



Oscillator



Group 8 :

Mahra: combined everything and provided diagrams.

Sumyya: Lectures 1 & 2, and real-life applications.

Swara: lecture 2&3.

Tanisha: Example questions.